

Setup Guide: Autonomous Beacon Return

Time Allocation: 90 Minutes

Challenge Overview

The robot must travel away from a home base, circle a distant pillar, and return autonomously to the starting point.

Environment & Arena Setup

- **Home Base:** Mark a clearly defined square (e.g., 50cm x 50cm) on the floor using brightly colored tape to represent the starting and ending home base for the robots.
- **The Pillar:** Place a single vertical pillar (such as a PVC pipe, a traffic cone, or a weighted cardboard box with a distinct color or visual marker) at a distance of 3 to 4 meters away from the home base.
- **Clearance & Boundaries:** Ensure there is a clear, unobstructed path of at least 2 meters around the pillar to allow robots to circle it safely without colliding with classroom furniture or other teams' equipment.
- **Lighting Conditions:** Stabilize the ambient lighting in the room as much as possible to ensure consistent sensor readings if teams are utilizing light or color sensors for navigation or line/pillar detection.